

Visual Outcome after Proton Beam Irradiation of Uveal Melanoma

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Abstract: Prognostic factors for visual loss following proton irradiation of uveal melanoma were evaluated for 440 eyes treated from 1975 to 1984, with visual acuity 20/200 or better before treatment. Analysis involved Kaplan-Meier survival curves and Cox proportional hazards analysis with visual outcome defined as worse than 20/200. Prognostic factors were tumor height: rate ratio (ratio of rate of visual loss for one category of the variable relative to the rate of visual loss for a reference category of that variable) of 5.26 (95% confidence interval, 2.66–10.39) for tumors greater than 5 mm compared to tumors 3.0 mm or less in height; distance of tumor from the optic disc and fovea: rate ratio 2.59 (1.63–4.11) for tumors 2DD or less from both the optic disc and fovea compared to those greater than 2 DD from these structures. Also predictive of visual loss were tumor location close to disc only, or close to fovea only, macular detachment, worse pretreatment vision, and higher radiation doses delivered to both the disc and fovea, and lens. Regression analysis using a visual acuity scale gave similar results. [Key words: multivariate analysis, prognostic factors, proton beam irradiation, tumor height, tumor location, uveal melanoma, visual outcome.] Ophthalmology 93:666–674, 1986

The major goal of radiation therapy for uveal melanoma is to destroy the tumor while preserving some vision and retaining the globe, without jeopardizing survival. Thus, the two important outcomes to evaluate are survival and visual function. Evidence to date suggests that there is no deleterious effect on survival in the early follow-up time after treatment when comparing proton irradiation to enucleation, but longer follow-up and additional studies are needed to confirm these findings.¹

Little is known concerning the risk factors for visual

loss after radiation therapy. A few reports have discussed visual results after radiation therapy.^{2–7} However, most of these studies involve a small number of patients and some do not account for the variable length of time from treatment to analysis of data. In the only report where risk factors were evaluated by actuarial methods, higher radiation dose to the disc and fovea, and larger tumor size were related to substantial visual loss after two to three years for patients treated with cobalt plaques.⁵ However, multivariate analysis was not performed to assess which variables were most important and which were independently related to visual loss.

Identification of factors related to poor visual outcome has important implications for decisions concerning choice of therapy and protocol development for a possible randomized controlled clinical trial comparing various treatments.

The purpose of this report is to identify which factors are important predictors of visual loss among patients with uveal melanoma treated by proton beam irradiation

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